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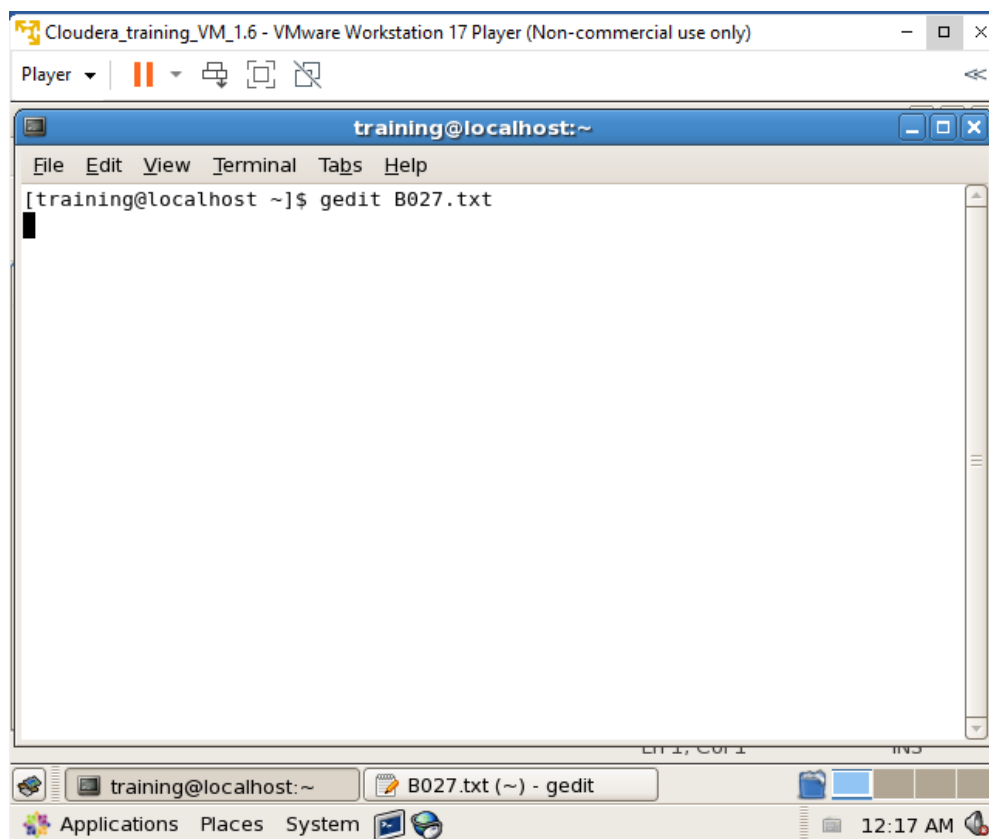
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EXPERIMENT 6

Aim: Execute PIG built-in commands and run pig scripts on HDFS.

Screenshots:

- 1) **create text/csv file on local user account and enter employee's details**





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```
training@localhost:~  
File Edit View Terminal Tabs Help  
[training@localhost ~]$ gedit B027.txt  
  
quit  
  
[training@localhost ~]$ cat B027.txt  
  
Eclipse  
training@localhost:~  
Applications Places System 12:18 AM
```

```
training@localhost:~  
File Edit View Terminal Tabs Help  
[training@localhost ~]$ gedit B027.txt  
  
quit  
  
[training@localhost ~]$ cat B027.txt  
123,abc,9.0,27  
234,def,8.5,30  
345,ghi,7.25,44  
678,jkl,8.4,60  
910,mno,9.5,28  
[training@localhost ~]$  
  
Eclipse  
training@localhost:~  
Applications Places System 12:21 AM
```



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```
training@localhost:~  
File Edit View Terminal Tabs Help  
[training@localhost ~]$ gedit B027.txt  
quit  
  
[training@localhost ~]$ cat B027.txt  
123,abc,9.0,27  
234,def,8.5,30  
345,ghi,7.25,44  
678,kl,8.4,60  
910,mno,9.5,28  
[training@localhost ~]$ hadoop fs -mkdir /test123  
[training@localhost ~]$ hadoop fs -put B027.txt /test123
```

```
[training@localhost ~]$ hadoop fs -put B027.txt /test123  
[training@localhost ~]$ hadoop fs -cat /test123/B027.txt  
123,abc,9.0,27  
234,def,8.5,30  
345,ghi,7.25,44  
678,kl,8.4,60  
910,mno,9.5,28  
[training@localhost ~]$
```

2) Copy stud1.txt file on HDFS

```
g/wordcount  
grunt> fs -mkdir /test123  
mkdir: cannot create directory /test123: File exists  
grunt> fs -mkdir /test234  
grunt> fs -put /home/training/B027.txt /test234/  
grunt> fs -cat /test234/B027.txt  
123,abc,9.0,27  
234,def,8.5,30  
345,ghi,7.25,44  
678,kl,8.4,60  
910,mno,9.5,28  
grunt>
```



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3) **Load the B027.txt to pig using the load command and store the result in student variable**

```
grunt> student = LOAD '/test234/B027.txt' USING PigStorage(',') AS (id:int,na  
me:chararray,cgpa:float,rollno:int);  
grunt> dump student  
2025-03-03 00:56:53,820 [main] INFO org.apache.pig.tools.pigstats.ScriptStat
```

dump student;

```
2025-03-03 00:57:16,250 [main] INFO org.apache.hadoop.mapreduce.lib.input.Fi  
leInputFormat - Total input paths to process : 1  
2025-03-03 00:57:16,250 [main] INFO org.apache.pig.backend.hadoop.execution  
engine.util.MapRedUtil - Total input paths to process : 1  
(123,abc,9.0,27)  
(234,def,8.5,30)  
(345,ghi,7.25,44)  
(678,jkl,8.4,60)  
(910,mno,9.5,28)  
grunt>
```

4) **Display separate columns from a given file.**

```
2025-03-03 01:12:20,081 [main] INFO org.apache.pig.backend.hadoop.execution  
engine.util.MapRedUtil - Total input paths to process : 1  
grunt> ds1 = foreach student generate id,cgpa;  
grunt> dump ds1;  
2025-03-03 01:12:31,799 [main] INFO org.apache.pig.backend.hadoop.execution  
engine.util.MapRedUtil - Total input paths to process : 1  
(123,9.0)  
(234,8.5)  
(345,7.25)  
(678,8.4)  
(910,9.5)  
grunt>
```

5) **Use order operator**

```
grunt> ds2 = order student by cgpa desc;  
grunt> dump ds2;
```



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```
engine.util.MapRedUtil - Total input paths to process : 1
(910,mno,9.5,28)
(123,abc,9.0,27)
(234,def,8.5,30)
(678,kl,8.4,60)
(345,ghi,7.25,44)
grunt> █
```

6) Use group

```
grunt> gr = group ds2 by name;
grunt> dump gr;
```

```
2025-03-03 01:21:19,688 [main]
engine.util.MapRedUtil - Total :
(abc,{(123,abc,9.0,27)})
(def,{(234,def,8.5,30)})
(ghi,{(345,ghi,7.25,44)})
(jkl,{(678,kl,8.4,60)})
(mno,{(910,mno,9.5,28)})
grunt>
```

7) Use filter operator

```
grunt> ds3 = filter student by name == 'def';
grunt> dump ds3;
```

```
engine.util.MapRedUtil - Total input paths to process : 1
(234,def,8.5,30)
grunt> █
```

8) Union operator

```
(123,abc,9.0,27)
grunt> Result = UNION ds3,ds4;
grunt> dump Result;
```

```
2025-03-03 01:20:00,470 [main]
engine.util.MapRedUtil -
(234,def,8.5,30)
(123,abc,9.0,27)
grunt> █
```



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9) **Store files on HDFS using the store command.**

```
Details at logfile: /home/training/pig_1740992896651.log
grunt> store Result into '/a6/' USING PigStorage(',');
2025-03-03 01:33:37 315 Finished TMRB ...
```

Output(s):
Successfully stored 2 records (30 bytes) in: "/a6"

Counters:
Total records written : 2
Total bytes written : 30

10) **Check the file has been copied**

```
grunt> fs -ls /a6/
Found 3 items
drwxr-xr-x  - training supergroup      0 2025-03-03 01:33 /a6/_logs
-rw-r--r--  1 training supergroup     15 2025-03-03 01:33 /a6/part-m-000000
-rw-r--r--  1 training supergroup     15 2025-03-03 01:33 /a6/part-m-000001
grunt>
```

11) **Display the result**

```
grunt> fs -cat /a6/part-m-000000
234,def,8.5,30
grunt>
```

Conclusion: We studied how to run the Pig command and Pig script using the Pig Hadoop ecosystem.